MIL-T-0037144B (DLA-DM)
15 February 1978
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12 May 1975

MILITARY SPECIFICATION

THEOPHYLLINE AND GUAIFENESIN CAPSULES

This limited coordination military specification has been prepared by the Defense Personnel Support Center, Defense Logistics Agency, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-T-37144A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers Theophylline and Guaifenesin Capsules.

1.2 Classification. The capsules covered by this specification shall be the following sizes (see 6.2):

Size 1 - Individually Sealed, 100s
Size 2 - 1000s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-285 - Capsules (for Medicinal Purposes).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Theophylline and Guaifenesin Capsules, and shall be in accordance with all applicable requirements of FED-STD-285, and as specified herein:

S2 Classification. Shall be Type II, Class 2.

S5.4 The following is added to this paragraph:

Each capsule shall contain 150 mg of anhydrous theophylline and 90 mg of guaifenesin, within the designated assay limits for the capsule. The capsule shall contain the active ingredients in a suspension of glycerin, polyethylene glycol 400, and polyethylene glycol 6000.

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The capsule shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of anhydrous theophylline and guaifenesin, when assayed as specified in 4.5.

Identity. The active ingredients in the finished capsule shall comply with the following identity tests:

Guaifenesin. The assay solution prepared as directed in 4.5.1.1 for guaifenesin shall exhibit a maximum absorbance at 276 ± 2 nm.

Theophylline, anhydrous. The assay solution prepared as directed in 4.5.1.2 for anhydrous theophylline, shall exhibit a maximum absorbance at 274 ± 2 nm.

If the alternate assay is employed (4.5.2), a suitable identity test for each active ingredient shall be performed.

pH. A solution of three (3) capsules per 100 ml of purified water shall have a pH of 4.20 to 6.40, at 25° C, when determined potentiometrically using the USP method.

S6.1 Ingredients. The following is added to this paragraph:

The anhydrous theophylline used in the manufacture of the finished capsules shall be in accordance with the tests, standards, and requirements of the USP and, in addition, shall comply with the following:

Particle size. The particle size of the anhydrous theophylline shall be such that a minimum of 98 percent shall pass through a No. 200 US Standard Sieve.

All other ingredients used in the manufacture of the finished capsules shall comply with S5.2.

S6.7.1 Classification of defects. Inspection for Defect #105 (Capsule empty), Table VI. Soft Capsules, shall be performed by weighing 125 capsules, cutting them open, and determining the weight of fill.

3.2 Expiration dating period. The expiration dating period for size 1 shall be not less than 24 months and the expiration dating period for size 2 shall be not less than 36 months.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The classification of defects as shown in FED-STD-285 shall be applicable.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests.

4.5.1 Sample preparation.

Cut ten (10) capsules in half and quantitatively transfer to a 500 ml volumetric flask. Add approximately 350 ml of purified water and shake until the contents of the capsules have dissolved. Add purified water to volume and mix well. Pipet 15.0 ml of the solution into a 125 ml separatory funnel, add about 15 ml of 1N sodium hydroxide solution and mix by swirling. Extract with three 25 ml portions and one 20 ml portion of chloroform. Drain each chloroform layer through a pledget of cotton in the stem of the separatory funnel into a 100 ml volumetric flask.

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Save the chloroform layer for the determination of guaifenesin and immediately add 3 ml of concentrated hydrochloric acid to the aqueous layer, mix by swirling and save this layer for the determination of theophylline.

4.5.1.1 Guaifenesin assay.

Using the chloroform extract, above, make to volume with chloroform and mix well. Pipet 15.0 ml into another 100 ml volumetric flask, make to volume with chloroform and mix well. Quantitatively and stepwise, prepare a solution of NF guaifenesin reference standard containing approximately 40 mcg per ml in chloroform. Concomitantly determine the absorbances of the sample and standard solutions at the maxima at about 276 and 310 nm in 1 cm cells with a suitable spectrophotometer using chloroform as the blank. Calculate the amount of guaifenesin as follows:

$$\text{mg per capsule} = \frac{(Au^{276} - Au^{310}) \times C \times 100 \times 500 \times 100}{(As^{276} - As^{310}) \times 15 \times 10 \times 15 \times 1000}$$

where Au is the absorbance of the sample at the designated wavelengths
As is the absorbance of the standard at the designated wavelengths
C is the concentration of the standard in mcg per ml.

$$\text{Percent of required amount} = \frac{\text{mg per capsule}}{90} \times 100$$

4.5.1.2 Theophylline, anhydrous, assay.

Use the acidified aqueous extract. Remove the cotton from the stem of the separatory funnel and extract the solution with 50 ml of Diethyl Ether, USP. Drain the aqueous (bottom) phase into a second 125 ml separatory funnel containing 25 ml of ether. Shake the second funnel for about 30 seconds and allow the layers to separate. Drain the aqueous layer into a 200 ml volumetric flask. Drain the ether layer in the second separatory funnel into the first separatory funnel; wash the second funnel with three 3 ml portions of ether and add the washings to the first separatory funnel. Extract with two, 25 ml portions of 0.1N hydrochloric acid. Allow the layers to separate and drain the aqueous layer into the same 200 ml volumetric flask. Add purified water to volume to the volumetric flask and mix. Pipet 5.0 ml of this solution into a 200 ml volumetric flask.

Quantitatively and stepwise prepare a solution of USP theophylline reference standard containing approximately 250 mcg of anhydrous theophylline per ml. Pipet 5.0 ml of this solution into a 200 ml volumetric flask.

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Concomitantly treat the sample and standard solutions as follows:

Add 25 ml 1N sodium hydroxide and dilute to volume with purified water. Mix and immediately determine the absorbances at the maxima at about 274 and 320 nm in 1 cm cells with a suitable spectrophotometer using purified water as a blank. Calculate the amount of anhydrous theophylline as follows:

mg anhydrous theophylline per capsule =

$$\frac{(Au^{274} - Au^{320}) \times C \times 500 \times 200}{(As^{274} - As^{320}) \times 10 \times 1000 \times 15}$$

where Au is the absorbance of the sample at the designated wavelength
As is the absorbance of the standard at the designated wavelength
C is the concentration of the standard in mcg per ml

$$\text{percent of required amount} = \frac{\text{mg per capsule} \times 100}{150}$$

4.5.2 Alternate assay. As an alternate to 4.5.1 the following assay procedure involving high pressure liquid chromatography may be used to determine theophylline and guaifenesin.

Equipment:

A suitable high pressure liquid chromatograph with a dual channel UV detector, a dual pen strip chart recorder, and a 20 µl sample volume loop injection valve. An automatic sampler may be used if available.

A suitable liquid chromatographic column (4 mm ID x 30 cm).

Reagents and Solutions:

Methanol, ACS reagent grade.

Potassium phosphate, monobasic (KH_2PO_4), ACS reagent grade.

10 percent w/v sodium hydroxide solution (aqueous), approximate.

pH 6.5 Buffer solution - Place 1.36 gram of KH_2PO_4 in a 1 liter volumetric flask and add 500 ml of water to dissolve the solid. Dilute to volume with water; then carefully adjust the pH to 6.5 with the 10 percent NaOH solution.

Mobile phase solvent - Mix 700 ml of the pH 6.5 buffer with 300 ml of methanol to give a 30 percent (v/v) methanol in buffer solution.

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Theophylline USP reference standard

Guaifenesin NF reference standard

Caffeine reference standard

Internal standard solution - Dissolve approximately 47 mg of caffeine in 1000 ml mobile phase solvent.

Reference standards stock solution - Weigh accurately about 150 mg of anhydrous theophylline reference standard and 90 mg of guaifenesin reference standard and place into a 200 ml volumetric flask. Add approximately 150 ml of pH 6.5 buffer and shake until the standards are completely dissolved. Dilute to volume with pH 6.5 buffer.

Reference standards working solution - Place a 5.0 ml aliquot of reference standards stock solution into a 100 ml volumetric flask and dilute to volume with internal standard solution.

Procedure:

Place five capsules into a 100 ml volumetric flask and add about 80 ml of pH 6.5 buffer. Heat the flask on the steam bath and swirl occasionally until the capsules are completely dissolved. This procedure should take about 10 minutes. A yellow cloudy solution is obtained. Cool the contents of the flask to room temperature with the aid of a cold water bath and then make up to volume with pH 6.5 buffer. Dilute this solution 1:10 by placing a 10.0 ml aliquot in a 100 ml volumetric flask and making up to volume with pH 6.5 buffer. Finally, place a 5.0 ml aliquot of this diluted solution into a 100 ml volumetric flask and make up to volume with internal standard solution. This sample solution is now ready for LC analysis.

Typical LC conditions:

Flow rate:	60 ml/hr
Temperature:	Ambient
Detector:	280 nm, both channels; Channel 1, 0.2 absorbance unit full scale (AUFS); Channel 2, 0.01 AUFS.
Recorder:	10 mv span both channels; Chart Speed = 0.1 in/min.

Allow the instrument to run until a fairly steady baseline is achieved (some drift on the 0.01 AUFS channel is acceptable) and then begin injecting the reference standards working solution until duplicate injections give R factors for both theophylline and guaifenesin that agree within $\pm$ 2 percent (relative). Inject samples in duplicate. Inject the standard solution after every second sample. The theophylline and internal standard peaks are read from channel 1 (0.2 AUFS) and the guaifenesin peak is read from channel 2 (0.01 AUFS).

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Approximate retention times:

Theophylline	5.5 min
Caffeine	7.5 min
Guaifenesin	11.0 min

NOTE: There is a small (detected on Channel 2) peak eluting at about 19.5 min which is due to the capsule shell or fill vehicle. Care must be taken in timing injections so that this residual peak from one injection will not overlap the peaks of interest from the subsequent injection.

Calculations:

Determine the R factors for each injection. Average the R values for the sample and the standard.

$$R_1 = \frac{\text{peak height theophylline}}{\text{peak height caffeine}}$$

$$R_2 = \frac{\text{peak height guaifenesin}}{\text{peak height caffeine}}$$

Theophylline:

$$\frac{R_1 \text{ sample}}{R_1 \text{ std.}} \times \text{Wt. of theophylline ref. std.(mg)} = \text{mg (anhydrous) theophylline per capsule}$$

$$\text{percent of required amount of theophylline} = \frac{\text{mg per capsule}}{150} \times 100$$

Guaifenesin:

$$\frac{R_2 \text{ sample}}{R_2 \text{ std.}} \times \text{Wt. of guaifenesin ref. std.(mg)} = \text{mg guaifenesin per capsule}$$

$$\text{percent of required amount of guaifenesin} = \frac{\text{mg per capsule}}{90} \times 100$$

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing one hundred capsules, individually sealed, or one bottle containing one thousand capsules, as specified, constitutes one unit of issue.

5.2.1.1 Immediate container.

5.2.1.1.1 Size 1. The immediate container for size 1 shall be a commercially available blister or strip packaging considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.1.1.2 Size 2. The immediate container for size 2 shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.1.2 Unit package. At the option of the contractor, a unit package may be furnished for size 2 capsules. If a unit package is furnished, each immediate container (bottle) shall be packaged as specified in 5.2.5 of PPP-C-186. A unit package shall be furnished for size 1 capsules. One hundred individually blister or strip packaged capsules shall be packaged in a box as specified in 5.2.5 of PPP-C-186.

5.2.1.3 Intermediate package. Ten unit packages containing size 1 capsules shall be packaged in an intermediate package as specified in 5.2.5 of PPP-C-186. Twelve units of issue of size 2 capsules shall be packaged in an intermediate package as specified in 5.2.9 of PPP-C-186.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

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5.3.1 Level A. Ten intermediate packages containing size 1 capsules or one intermediate package containing size 2 capsules shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ten intermediate packages containing size 1 capsules shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186. Unless otherwise specified, no overpacking will be required for size 2 capsules. The intermediate package shall serve as the exterior container.

5.3.3 Level C. The packaged commodity shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) for size 2 only (1000s): the expiration date.

5.4.1.2 Unit package. Each unit package shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.
(c) for size 1 only (100s): the expiration date.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied for size 2, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

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5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-00-118-2496	THEOPHYLLINE AND GUAIFENESIN CAPSULES, Individually Sealed, 100s	Size 1
6505-00-764-3366	THEOPHYLLINE AND GUAIFENESIN CAPSULES, 1000s	Size 2

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
DLA-DM

Project No. 6505-2007

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-0037144B (DLA-DM) THEOPHYLLINE AND GUAIFENESIN CAPSULES

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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